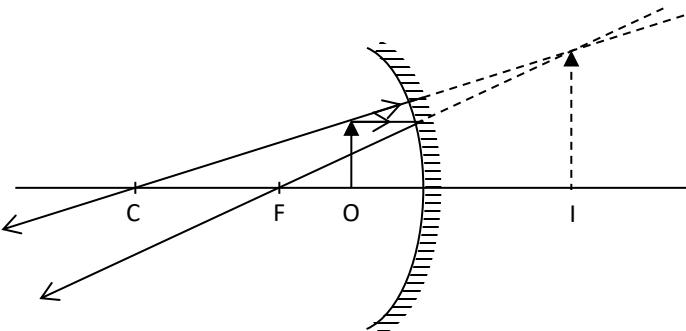
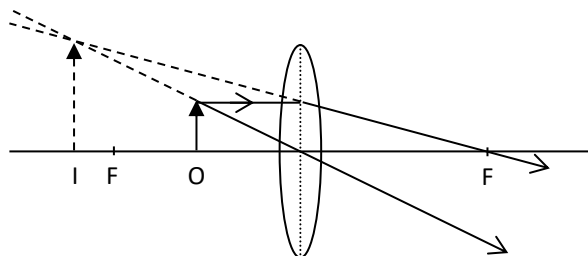


ANSWER CHAPTER 6 CH

NO	SCHEME	REMARKS
1	a) Concave mirror b) 	
2	a) Given $u = 10 \text{ cm}$, $r = 30 \text{ cm}$, concave mirror $\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \text{ and } f = \frac{r}{2}$ $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ $\frac{1}{v} = \frac{2}{r} - \frac{1}{u}$ $\frac{1}{v} = \frac{2}{30} - \frac{1}{10}$ $v = -30 \text{ cm}$ Position of image is -30 cm from the pole of the concave mirror b) Characteristics of the image are virtual, upright, and magnified.	
3	Given $u = 30 \text{ cm}$, $v = -5 \text{ cm}$, virtual image $\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \text{ and } f = \frac{r}{2}$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{2}{r} = \frac{1}{30} + \frac{1}{(-5)}$ $r = -12 \text{ cm}$ This is a convex mirror.	
4	a) Given $m = -2$, virtual image, $u = 10 \text{ cm}$	

	$m = \frac{v}{u}$ $v = mu$ $v = -2(10)$ $v = -20\text{cm}$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{f} = \frac{1}{10} + \frac{1}{(-20)}$ $f = 20\text{ cm}$ b) Concave mirror	
5	i. $f = \frac{r}{2} = 15\text{ cm}$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{15} = \frac{1}{u} + \frac{1}{-20}$ $u = 8.57\text{cm}$ ii. $m = \frac{v}{u} = \frac{h_i}{h_o}$ $\frac{-20}{8.57} = \frac{h_i}{12}$ $h_i = 28\text{ cm}$ iii. image is virtual and upright	
6	a) $u = v = 12\text{ cm}$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{f} = \frac{1}{12} + \frac{1}{12}$ $f = 6\text{ cm}$ b) concave mirror or converging mirror	
7	Image formation of a convex lens for an object that produces an upright and magnified image.	



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Given $h_o = 1.5 \text{ cm}$, $u = 13 \text{ cm}$, $f = 13.5 \text{ cm}$, converging lens

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{13.5} - \frac{1}{13}$$

$$v = -351 \text{ cm}$$

Position of the image is 351 cm from the centre of the lens on the same side as the object.

Characteristics of the image produced are virtual, upright, and magnified.

From the definition of magnification,

$$m = \frac{h_i}{h_o} = \frac{v}{u}$$

$$h_i = \frac{v}{u} h_o$$

$$h_i = \frac{351}{13} (1.5)$$

$$h_i = 40.5 \text{ cm}$$

Height of the image is 40.5 cm.

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$$m = \frac{h_i}{h_o} = \frac{20}{10} = \frac{v}{u} = 2$$

$$m = \frac{v}{u} = 2$$

$$v = -mu = (-2)(30) = -60 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

	$\frac{1}{f} = \frac{1}{30} + \frac{1}{-60}$ $f = 60 \text{ cm}$	
10	$\frac{1}{f} = (n - 1) \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$ $\frac{1}{f} = (1.5 - 1) \left(\frac{1}{20} - \frac{1}{-20} \right)$ $f = 20 \text{ cm}$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{20} = \frac{1}{10} + \frac{1}{v}$ $v = -20 \text{ cm}$	